

Concordia Le Gym Application

SOEN 6751 – Human Computer Interface Design

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I. INTRODUCTION

Many students at Concordia University face significant challenges in managing their gym activities at *Le Gym* due to inefficient and fragmented systems. Booking fitness classes, scheduling personal training sessions, and staying informed about gym events often require multiple steps, leading to frustration and missed opportunities. Additionally, the lack of real-time updates on class availability, gym occupancy, and personalized fitness tracking makes it difficult for students to plan and optimize their workouts. Without an integrated platform, students must rely on different tools and manual efforts to manage their fitness routines, which discourages engagement and creates barriers to maintaining an active lifestyle.

Our project aims to address these accessibility and efficiency issues by developing a mobile application that centralizes all gym-related tasks. This app will allow students to seamlessly book fitness classes, schedule personal training sessions, and rent lockers, all within a single, user-friendly interface. Real-time updates on gym occupancy and class availability will enable users to make informed workout decisions, reducing wasted time and frustration.[1] The app will also feature an integrated payment system for managing memberships, class payments, and locker rentals, streamlining financial transactions and eliminating the need for cumbersome manual processes. Additionally, a personalized fitness tracking feature will allow users to log workouts, monitor progress, and receive tailored recommendations to support their fitness goals.

Through this project, we expect to answer key questions about gym accessibility and student engagement. Specifically, we aim to explore how gym management tools can be improved to enhance usability and efficiency, what features best address students' needs for fitness tracking and real-time information, and how an integrated platform can encourage greater participation in fitness activities. By addressing these questions, we hope to develop a solution that not only resolves existing issues but also fosters a more active and health-conscious university community.

Our motivation for exploring this issue stems from the frustration among Concordia students regarding the gym's current management system. Many students struggle with booking classes due to unclear scheduling tools, miss out on important events because of a lack of notifications, or waste time visiting the gym only to find it overcrowded. By understanding these concerns and analyzing common pain points, we see an opportunity to create a seamless, efficient, and engaging solution that enhances the overall fitness

experience. Beyond solving a practical problem, this project is also an opportunity for our team to gain hands-on experience in software development, UI/UX design, and user-centered application development.

We strongly believe this is a problem worth addressing because the current gym management experience is outdated and does not align with the digital expectations of students.[2] The lack of a centralized system forces students to navigate multiple tools when planning workouts. Many universities have already adopted mobile-first gym management solutions, demonstrating the effectiveness of such platforms in increasing student participation and satisfaction. By developing an all-in-one mobile application tailored to the needs of Concordia students, we aim to bridge the gap between gym services and user expectations, ultimately promoting a healthier, more connected, and more engaged student community.

II. USER TASKS AND GOALS

Our mobile application is designed to enhance gym accessibility, efficiency, and engagement by providing an intuitive platform that streamlines all gym-related tasks. Users will be able to manage their fitness activities with ease, reducing the friction associated with traditional gym management systems. The app will empower students to efficiently plan their workouts, track progress, and stay informed about gym updates, ensuring a seamless fitness experience.

A. User Tasks

1) Class & Personal Training Bookings

Users will be able to check the availability of fitness classes and personal training sessions, schedule their bookings, and manage cancellations or rescheduling as needed. The system will provide a real-time calendar interface, allowing students to select their preferred sessions based on instructor availability, class capacity, and their own fitness goals. Automatic reminders and notifications will ensure that users do not miss their scheduled workouts.[3]

2) User Registration & Profile Management

To personalize the gym experience, students will be able to create user profiles where they can input their fitness goals, workout preferences, and experience levels. The app will use this information to provide class recommendations, suggest training programs, and track progress over time. Users can update their profiles at any time to refine their fitness journey.[3]

3) *Workout Tracking & Fitness Progress Monitoring*

Users will be able log their workouts, track performance metrics such as calories burned, repetitions, and workout duration, and receive insights based on their exercise history. The app will generate progress reports and visual analytics, helping users set new fitness goals and stay motivated. Additionally, integration with wearable fitness devices and mobile health apps could enhance data tracking capabilities.[4]

4) *Gym Occupancy Monitoring*

To optimize their workout schedules, users will be able to check real-time gym occupancy levels before visiting. This feature will display peak and off-peak hours, helping students avoid overcrowded gym sessions and better plan their workouts. The system may also suggest alternative time slots when the gym is less busy.[5]

5) *Real-Time Notifications & Updates*

The app will send users timely alerts regarding class reminders, last-minute schedule changes, gym maintenance closures, and special fitness events. Push notifications will ensure that students receive critical updates in real time, minimizing inconvenience and missed opportunities.[6]

6) *Usability-Focused Payment System*

Users will be able to perform transactions for class bookings, memberships, and locker rentals through a secure and simplified payment gateway. The system will support multiple payment methods, including credit/debit cards, mobile wallets, and university-linked payment systems. A clear transaction history will allow users to track payments and manage their gym-related expenses effortlessly.[7]

7) *Direct Communication with Gym Staff*

To enhance customer support and user engagement, the app will feature a built-in chat system where users can ask questions, request assistance, and provide feedback directly to gym staff. This will streamline communication and improve responsiveness to student needs, making gym interactions more efficient and user-friendly.[6]

8) *Locker Rental System*

Students will be able to check locker availability, reserve lockers, and manage their rental status through the app. The system will provide real-time updates on available lockers and notify users about upcoming rental expirations, ensuring smooth locker management without the need for in-person inquiries.[1]

B. *User Goals*

1) *Enhanced Gym Accessibility*

Users want to reduce the barriers to gym participation by interacting with an intuitive and user-friendly platform. Instead of navigating fragmented systems, they will have access to a streamlined interface where they can effortlessly book classes, schedule personal training sessions, and manage their gym memberships. The app ensures that fitness activities are accessible to all students, regardless of their familiarity with existing gym management tools.[3]

2) *Optimized Time Management*

Students want to efficiently plan their workouts around their academic and personal schedules. With real-time updates on gym occupancy and class availability, users can make informed decisions about when to visit Le Gym, avoiding

peak hours and ensuring that they maximize their time. The system will also provide notifications and reminders for upcoming sessions, reducing the likelihood of missed workouts and last-minute scheduling conflicts.[4]

3) *Seamless Interaction with Gym Facilities*

Users want a centralized platform that consolidates all gym-related tasks into one place. From booking and payments to locker rentals and fitness tracking, students will no longer need to rely on multiple systems or manual processes. This level of integration simplifies gym engagement, making it easier to stay consistent with their fitness routines.[5]

4) *Improved Fitness Tracking and Goal Setting*

Students want to set personalized fitness goals and monitor their progress over time. By logging workouts, tracking performance metrics, and receiving tailored workout recommendations, users can stay on top of their fitness journey. Progress reports and analytics will provide insights into their achievements, helping them stay motivated and make data-driven adjustments to their exercise routines.[6]

5) *A More Engaging & Motivating Gym Experience*

Maintaining fitness motivation can be challenging, and users want to stay engaged with their workouts through interactive features. Achievement tracking, milestone notifications, and personalized reminders will encourage consistency and reinforce positive habits. The app will also offer encouragement through goal-based progress indicators, helping users maintain a sense of accomplishment.[7]

6) *Stronger Communication & Support*

Users want quick and efficient communication with gym staff for assistance, inquiries, and feedback. With a built-in messaging system, students can ask questions, request support, or report issues directly through the app. This direct communication channel ensures smoother gym experience and improves responsiveness to user needs.[6]

C. *Intended Users, Their Needs, and Solutions*

Our primary users are Concordia University students who visit Le Gym and require a more efficient and engaging way to manage their fitness activities. The app is designed to accommodate a diverse range of users with varying fitness levels, goals, and technological familiarity, ensuring an inclusive and user-friendly experience.[1]

1) *User Groups and Their Needs, Preferences, and Limitations*

a) *Beginner Gym Users*

- *Needs:* Guidance on how to start, class recommendations, and easy access to trainer support.
- *Preferences:* Simple, intuitive interfaces with clear instructions, structured workout plans, and recommendations.
- *Limitations:* Lack of knowledge about workout routines, class schedules, and proper gym usage.
- *Challenges:* Beginners often feel overwhelmed by gym equipment and unsure of how to structure workouts effectively.
- *Solutions:* The app will provide beginner-friendly workout plans, a class recommendation system,

and direct access to trainers for guidance. Step-by-step instructional videos and structured onboarding will ease their transition into fitness.

b) Regular Fitness Enthusiasts

- *Needs:* Advanced workout tracking, goal-setting tools, and detailed scheduling options.
- *Preferences:* Efficiency, flexibility, and performance tracking with clear progress indicators.
- *Limitations:* Limited ability to manually track workouts and adjust routines dynamically.
- *Challenges:* Managing training schedules, tracking progress manually, and staying consistent with workout plans.
- *Solutions:* The app will feature automated workout logs, personalized fitness goals, and performance analytics. Users can set training schedules, receive reminders, and track their improvement over time.

c) Advanced Users

- *Needs:* Customizable workout plans, in-depth performance analytics, and integration with wearable fitness devices.
- *Preferences:* High levels of customization, real-time insights, and data-driven fitness optimization.
- *Limitations:* Limited access to advanced tracking features, specialized training plans, and gym performance insights.
- *Challenges:* Lack of advanced tracking tools, difficulty accessing specialized training resources, and the need for performance benchmarking.
- *Solutions:* The app will support wearable fitness device integration, goal-based performance metrics, and customizable workout plans, allowing users to fine-tune their training based on detailed analytics.

2) Challenges Users Might Face and How the App Addresses Them

a) Tracking Workout Progress

- *Challenge:* Users struggle to manually track their workout progress, making it difficult to measure improvement over time.
- *Solution:* The app automatically logs workouts, sets fitness goals, and provides detailed performance insights through progress graphs and summaries.

b) Difficulties in Booking Classes or Personal Training

- *Challenge:* The existing gym booking system is inefficient, leading to missed opportunities for classes and personal training sessions.
- *Solution:* The app will feature a real-time booking system where users can check availability, reserve

spots, and receive automated reminders for their upcoming sessions.

c) Complex Payment Processes

- *Challenge:* Membership renewals, class payments, and locker rentals currently involve cumbersome and fragmented payment methods.
- *Solution:* The app will integrate a secure, user-friendly payment gateway that allows students to handle all transactions in one place, with multiple payment options and automated receipts.

d) Limited Communication with Gym Staff

- *Challenge:* Users often struggle to get quick responses to their inquiries about gym policies, personal training sessions, or equipment availability.
- *Solution:* A built-in chat feature will allow users to directly communicate with gym staff for support, feedback, and quick assistance.

e) Lack of Awareness of Gym Occupancy

- *Challenge:* Students may arrive at the gym during peak hours without knowing how crowded it is, leading to a frustrating experience.
- *Solution:* A live occupancy tracking feature will display real-time gym capacity data, helping users plan their visits during less busy hours.

3) Why These Tasks and Goals Matter to Users

Each of these user needs and solutions is critical to enhancing the overall gym experience. Students want to spend less time managing their workouts and more time focusing on fitness. By addressing pain points such as inefficient booking, lack of real-time information, and poor workout tracking, the app ensures that users can make informed decisions, stay motivated, and achieve their fitness goals more effectively. Ultimately, the app will transform the way students interact with *Le Gym*, making it a more accessible, structured, and rewarding environment for all.

III. METHODOLOGY OVERVIEW

D. Core System Functionality

Our gym management system will provide a centralized, user-friendly platform that enhances accessibility, scheduling, fitness tracking, and communication for Concordia students at *Le Gym*. The app will streamline gym interactions through the following functionalities:

- *Seamless Class & Training Bookings:* Users can check availability, reserve spots, and receive real-time updates on fitness classes and personal training sessions.
- *User Profiles & Fitness Tracking:* Personalized dashboards will enable students to log workouts, track progress, and receive tailored recommendations.
- *Real-Time Gym & Facility Updates:* Live occupancy data and equipment availability

insights help users plan efficient workout sessions.

- *Integrated Payment & Membership Management:* Secure transactions for class bookings, membership renewals, and locker rentals simplify the user experience.
- *Locker Reservations & Management:* A digital system for checking locker availability, booking rentals, and receiving expiration reminders.
- *Instant Notifications & Alerts:* Automated reminders for upcoming bookings, gym announcements, and schedule changes keep users informed.
- *Direct Communication with Gym Staff:* A built-in messaging feature allows quick access to support and inquiries.

E. Usability Approach and Design Strategy

Creating an intuitive and efficient gym management application requires a *user-centered design* (UCD) approach, ensuring that the system is built around the actual needs, behaviors, and expectations of Concordia students. *User-centered design* is an iterative process that prioritizes usability by actively involving users throughout the design and development phases. This approach helps create an application that is accessible, efficient, and enjoyable to use, catering to students with varying fitness levels and technological familiarity.[8]

A key aspect of our design strategy is adhering to *Jakob Nielsen's 10 Usability Heuristics for User Interface Design*; a set of fundamental principles that guide the development of user-friendly digital systems. These heuristics serve as best practices to enhance usability, prevent frustration, and create a seamless interaction experience.[9]

a) Jakob Nielsen's 10 Usability Heuristics and Their Role in Our Application

1. *Visibility of System Status:* Users should always be informed about what is happening within the system through real-time feedback, progress indicators, and notifications (e.g., booking confirmations, gym occupancy updates).
2. *Match Between System and the Real World:* The interface should use familiar terms, symbols, and workflows that align with how students naturally think about gym activities (e.g., clear labels like "Book Class" instead of "Submit Request").
3. *User Control and Freedom:* Users should be able to easily cancel, undo, or modify actions, such as rescheduling a class or reversing an accidental gym membership renewal.
4. *Consistency and Standards:* The application should follow established design conventions, so users do not need to learn new interactions (e.g., standard navigation patterns, uniform button styles, and consistent terminology).
5. *Error Prevention:* The system should help users avoid mistakes by providing warnings,

confirmations, and input validation (e.g., preventing duplicate bookings or incorrect payment details).

6. *Recognition Rather Than Recall:* Users should not have to memorize functions or navigation paths; frequently used features like class bookings, locker rentals, and fitness tracking should be easily accessible.
7. *Flexibility and Efficiency of Use:* The app should cater to both beginners and experienced gym-goers by offering default fitness plans for new users and customization options for advanced users.
8. *Aesthetic and Minimalist Design:* The interface should remain clean, uncluttered, and focused on key actions, avoiding unnecessary elements that distract from core functionalities.
9. *Help Users Recognize, Diagnose, and Recover from Errors:* When errors occur, users should receive clear and helpful messages (e.g., "Payment Failed – Please Check Your Card Details" rather than vague errors like "Transaction Error").
10. *Help and Documentation:* A help center, FAQs, and chat support should be available to assist users without overwhelming them with excessive instructions.

b) Applying UCD and Heuristics to Our Design Strategy

By integrating *User-Centered Design* principles with *Nielsen's Usability Heuristics*, our approach will ensure that the gym management application:

1. Is built based on real student needs and challenges through research and feedback loops.
2. Provides a smooth and intuitive experience with easy-to-navigate workflows and meaningful interactions.
3. Prevents common frustrations by offering visibility, flexibility, and error handling mechanisms.
4. Encourages student engagement by making gym participation effortless, motivating, and rewarding.

With this foundation, we will follow the *Double Diamond Design Model* to systematically discover, define, develop, and deliver a highly usable gym management system.

IV. DOUBLE DIAMOND DESIGN PROCESS

To ensure a seamless and user-friendly experience, our design approach will follow the *Double Diamond Design Model*, a structured framework focused on problem-solving, iterative refinement, and user-centered innovation. This approach consists of four key phases—*Discover*, *Define*, *Develop*, and *Deliver*—allowing us to design an intuitive, efficient, and accessible gym management application. Our usability principles will be rooted in simplicity, efficiency, accessibility, consistency, and feedback-driven design, ensuring that all user interactions are smooth, engaging, and purposeful.[10]

F. Discover Phase

To build a highly functional and user-friendly gym management application, we began by deeply researching user behavior, common pain points, and system inefficiencies at Le Gym. Given the scope and limitations of the project, no direct surveys were conducted; however, we analyzed public user reviews from Google, examined the current Le Gym webpage, and compared it to other university fitness centers to identify usability barriers and areas for improvement.

1) User-Centered Research

Our research focused on key challenges related to gym accessibility, class booking, fitness tracking, and payments. Analyzing *Google Reviews* (3.8 average rating) revealed recurring complaints, including:

- **Overcrowding:** Many users expressed frustration over the gym being consistently packed, especially during peak hours. As one reviewer stated: *"Barely worth it even with the cheap student price. Consistently overcrowded for most of the day, unless you want to do nothing but cardio. There aren't nearly enough machines or benches to go around."*
- **Outdated Equipment:** Multiple reviews highlighted that the gym's equipment is old and in need of maintenance or replacement.
"The seat height adjustment on all the cable machines is broken."
"The gym has outdated equipment and lacks machines like back rows or shoulder presses."
- **Lack of Real-Time Information:** Users struggle with gym occupancy levels and equipment availability, often arriving at the gym only to find it too crowded to work out effectively.
- **Website and Booking Frustrations:** The *Le Gym* web page is a subsection of the overall Concordia website, making navigation difficult. Membership registration, class schedules, and bookings require multiple steps and searches, leading to a frustrating experience.

2) Analysis of Existing Systems

While Le Gym does provide affordable fitness facilities, it lacks a dedicated digital platform for seamless interaction. The current system does not provide real-time occupancy updates, streamlined class bookings, or an efficient payment system, which results in:

- A poor scheduling experience requires students to search through multiple sections to find class information.
- A fragmented payment process with unclear steps for membership registration, renewals, and locker rentals.
- Limited direct communication with gym staff, requiring students to rely on in-person visits or emails for inquiries.

3) Data-Driven Insights & Areas for Improvement

From our research, we identified several usability barriers and improvement opportunities, which our app will address:

- **Real-Time Occupancy Monitoring:** Allowing students to check gym crowd levels before

visiting, helping them plan workouts more efficiently.

- **Streamlined Class Booking System:** A centralized, one-tap booking system to eliminate confusion and reduce missed opportunities.
- **User-Friendly Membership and Payment Integration:** Simplifying registration, renewals, and locker rentals into a single, clear process.
- **Improved Workout Tracking & Fitness Guidance:** Offering customized workout recommendations and tracking tools for beginners and advanced users.
- **Direct Communication with Gym Staff:** Implementing in-app chat support for quick answers and guidance.

4) Usability Principles Applied

To ensure an improved user experience, our design approach follows key usability principles:

- **User-Centered Design:** Prioritize real user needs through research and validation.
- **Accessibility & Inclusivity:** Design an interface that accommodates beginners, fitness enthusiasts, and advanced users.
- **Discoverability & Simplicity:** Reduce complexity, making essential features (e.g., booking, occupancy tracking, payments) immediately accessible.

G. Define Phase

After identifying key user challenges and needs in the Discover Phase, we now focus on defining the core features and structuring user interactions to ensure an intuitive and efficient gym management application. This phase includes persona development, user flow mapping, and prototyping to refine the application's usability and functionality.

1) User Personas & Empathy Maps

To ensure our app effectively serves the diverse student population at Concordia, we developed three personas with empathy maps, each representing different fitness levels, backgrounds, and technical proficiency. These personas help us understand distinct user behaviors, goals, and frustrations, ensuring our design is inclusive and user centric.[10]

2) Persona 1: Wei Zhang – The Newcomer

- Age: 19
- Origin: China
- Program: Year 1 Computer Science
- Fitness Level: Beginner
- Tech Proficiency: High

a) Background & Needs

Wei is new to Canada and the Concordia campus, interested in playing soccer, but has never been to a gym before. His primary needs include:

- Multilingual support to overcome the language barrier.
- Step-by-step guidance on gym services and equipment.
- An easy way to book classes and personal training sessions.
- Seamless integration with his school schedule.

b) *Frustrations*

- Confusing class scheduling and gym accessibility.
- Difficulty finding available equipment.
- Locker accessibility issues.
- Preferred payment method (WeChat) not supported.

c) *Empathy map*

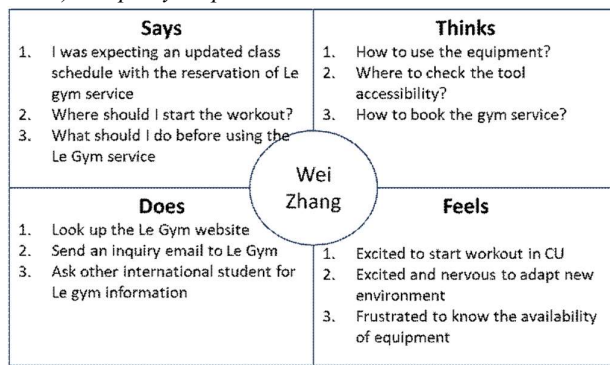


Figure 1: Persona 1 Empathy Map

3) *Persona 2: Gabrielle Dubois – The Enthusiast*

- Age: 21
- Origin: Canada
- Program: Year 3 Economics
- Fitness Level: Regular Gym-Goer
- Tech Proficiency: Medium

a) *Background & Needs*

Gabrielle has over two years of gym experience, is familiar with the campus, and has a gym membership outside Concordia but prefers *Le Gym* for convenience. She is looking for better gym management tools that will help her stay consistent with her workouts.

- Tracking progress and workout history.
- Reserving gym equipment and locker space.
- Finding gym partners and attending group classes.
- A more intuitive class booking system.

b) *Frustrations*

- Difficulty tracking past gym activities.
- Locker availability issues.
- Hard to find a regular workout partner.

c) *Empathy map*

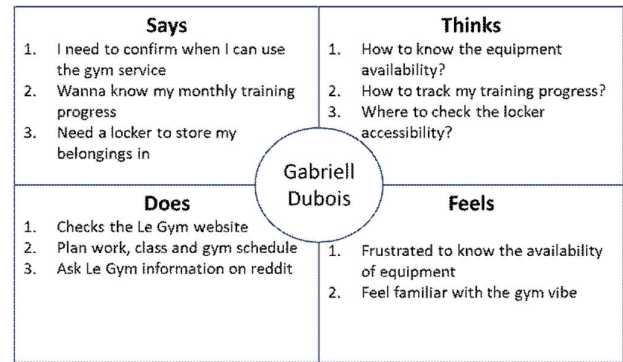


Figure 2: Persona 2 Empathy Map

4) *Persona 3: James Jones – The Athlete*

- Age: 27
- Origin: UK
- Program: PhD Year 2 Biology
- Fitness Level: Professional Athlete
- Tech Proficiency: High

a) *Background & Needs*

James was an athlete before starting his PhD and is a teaching assistant who wants to integrate gym sessions into his busy schedule. He is considering joining *Le Gym* but wants to ensure it meets his needs.

- Advanced training tracking features.
- Real-time gym availability monitoring.
- Flexible booking for equipment and personal coaching.
- Balancing gym time with academic workload.

b) *Frustrations*

- Inconsistent gym scheduling affecting research workload.
- Limited tracking tools for advanced athletes.
- Lack of specialized training resources.

c) *Empathy map*

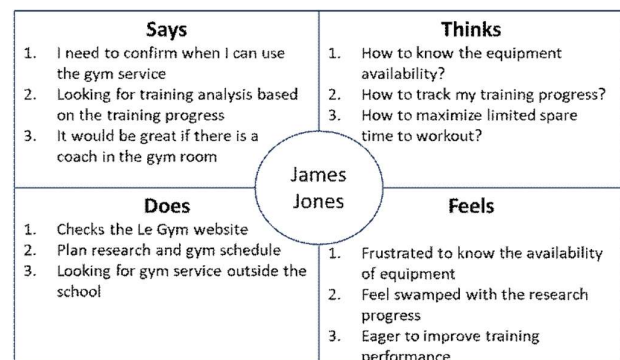


Figure 3: Persona 3 Empathy Map

5) *User Flow Diagrams & Journey Mapping*

We designed detailed flow diagrams to map out key user interactions, ensuring that every task within the app is efficient, intuitive, and error-free. These workflows are designed to minimize cognitive load and provide a structured, logical experience for students at different fitness levels.[10]

a) User Registration & Profile Management

A streamlined onboarding process that helps users set up profiles, fitness goals, and preferred workout schedules.



Figure 4: User Registration & Profile Management

b) Class Booking

A simple, real-time booking system allowing users to filter by time, instructor, and availability, with automated reminders.

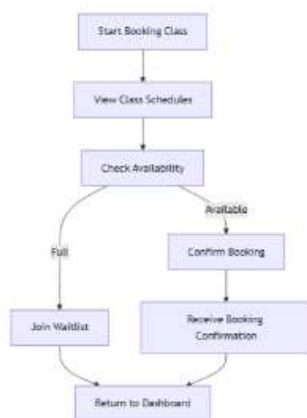


Figure 5: Class Booking

c) Reservation of Gym Equipment & Locker

Users can check equipment availability, reserve machines, and secure locker space before arriving.

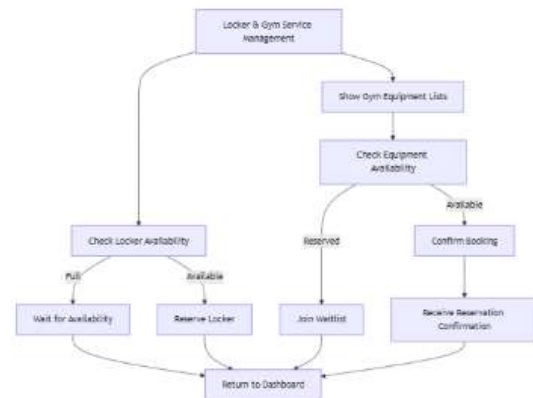


Figure 6: Reservation

d) Coach or Personal Training Booking

A booking system for one-on-one sessions with trainers, customized to personal fitness goals.



Figure 7: Coach Booking

e) Workout Tracking & Fitness Progress

Enables users to log workouts, track performance metrics, and receive fitness recommendations.

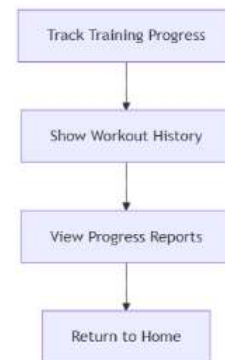


Figure 8: Workout Tracking

f) Payment System

Supports seamless transactions for memberships, locker rentals, and class fees, integrating multiple payment options.

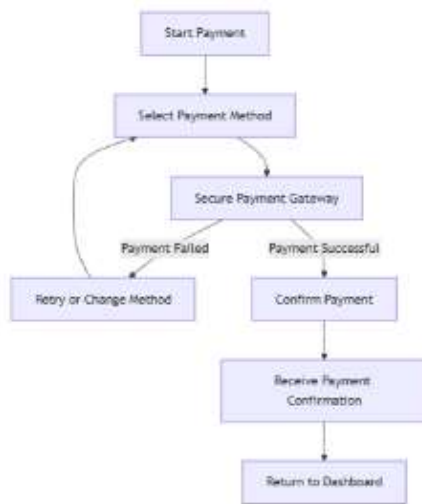


Figure 9: Payment System

g) Direct Communication with Gym Staff

A built-in chat feature for quick inquiries, assistance, and feedback.

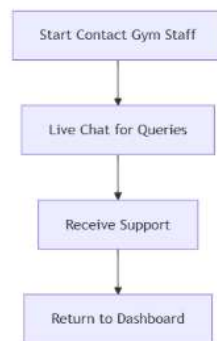


Figure 10: Communication with staff

h) Gym Occupancy Monitoring

Displays real-time crowd levels, allowing users to plan gym visits efficiently.

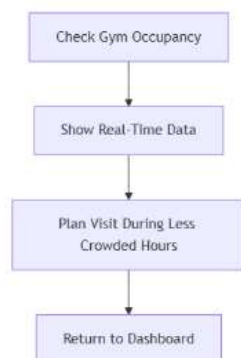


Figure 11: Gym Occupancy

6) User Journey Map

A user journey map was created to visualize the end-to-end experience of different personas. It highlights how users navigate through the app, from registration to booking workouts and tracking progress. By analyzing each step, we identified friction points and opportunities to enhance efficiency and usability. [FIGURE 12] [10]

7) Usability Principles Applied

To ensure a smooth and engaging user experience, our design strategy follows core usability heuristics:

- **Consistency & Familiarity:** Using standard UI conventions (e.g., familiar icons, clear labeling) for predictable interactions.
- **Task Efficiency:** Minimizing steps required to complete core tasks (e.g., booking a class in two taps).
- **Error Prevention:** Implementing confirmation prompts, warnings, and real-time validation to reduce mistakes.
- **Discoverability & Simplicity:** Designing workflows that make essential features immediately accessible, reducing the learning curve for new users.
- **User Control & Freedom:** Ensuring that users can easily modify bookings, cancel reservations, and undo actions to prevent frustration.
- **Visibility of System Status:** Planning for real-time updates on gym occupancy, class availability, and payment confirmations to keep users informed.

Since no prototypes have been created yet, these usability principles will serve as the foundation for future prototyping and iterative improvements in the upcoming Develop Phase.

By defining user personas, designing structured workflows, and optimizing usability, we ensure that our application meets the varied needs of Concordia students. Whether users are beginners, regular gym-goers, or advanced athletes, the app provides an intuitive, efficient, and enjoyable fitness management experience. This structured Define Phase allows us to move forward with prototyping and usability testing in the Develop Phase, ensuring that our application remains user-friendly and highly functional.

H. Develop Phase

With a well-defined structure in place, the next step is to develop, test, and refine the gym management application to ensure it meets usability standards and provides an efficient user experience. This phase involves prototyping, usability testing, and iterative improvements to enhance functionality, responsiveness, and ease of use.

At this stage, no prototypes have been created yet; however, the planned approach focuses on progressive development, feedback-driven refinement, and a mobile-first design strategy to ensure the app is intuitive and accessible to all users.

1) Prototyping & User Testing Approach

	Discovery			Registration			Pre-Workout					During Workout					Post-Workout			
Touchpoints	Concordia Le Gym Website			Concordia Le Gym Website		In person	Concordia Le Gym Website					Front Desk	Le Gym fitness centre		Front Desk	Concordia Le Gym Website				
Actions	Find Le Gym info	Check memberships options	Check class & equipment options	Create user profile	Set fitness goals	Payment	Plan visit	Check Gym Occupancy	Reserve Locker	Book class/ coach/ equipments	Plan workout routine	Pack personal belongings (towel)	Check in	Use Locker	Use equipment	Attend class/coach/ start raining	Record workout progress	Review training progress	Make reservation	Provide Feedback
Pain Points	Simplified info	/	No real-time availability	Cannot personalize training goal while creating the account		Online payment security concerns	Too crowded	No real-time availability	Not available	No real-time data	Didn't add to the schedule	/	/	/	Too crowded	Too crowded	No comprehensive progress	Not available	Not available	/
Emotions	😊	😐	😐	😐	😊	😊	😐	😐	😐	😐	😊	😊	😊	😊	😊	😊	😊	😊	😊	😊
Opportunities	Provide real-time class & equipment availability Provide FAQ section about Le Gym info			Allow user to set their training goals and preferences New Usability-Focused Payment System Provide real-time occupancy			Provide real-time occupancy, class, coach availability data Provide online locker booking service					Provide real-time occupancy and equipment availability data Allow reservation of the class and coach Provide a full training record in their user profile				Provide comprehensive training history Provide class, equipment and coach reservation service				

Figure 12: Journey Map

The development process will follow an iterative testing cycle, allowing us to identify usability issues early and refine the interface accordingly.

a) Low-Fidelity Prototypes

- **Paper wireframes** were initially sketched to outline navigation structures and key user flows.
- **Balsamiq wireframes** were then created to structure layout and test usability in a low-fidelity environment. [FIGURE 13]
- Focused on layout, structure, and core interactions rather than aesthetics.

b) High-Fidelity Prototypes

- Polished high-fidelity prototypes were designed using Figma, integrating the full visual identity, branding, and UI components for a seamless mobile experience. [FIGURE 13]
- These designs will mimic the final user experience, allowing for more detailed usability testing.
- Aesthetics, animations, and accessibility improvements will be refined based on user feedback.

c) Usability Testing & Refinement

- A/B Testing: Comparing different interface versions to determine the most intuitive design.
- Heuristic Evaluations: Using usability heuristics to assess potential usability issues.
- User Testing Sessions: Observing real users as they navigate the app to identify pain points and areas for improvement.

d) Successful App Development

We successfully developed the app using React Native, following the Figma design precisely. Each component was implemented to mirror the prototype in terms of layout, styling, navigation, and interactivity. Special attention was given to:

- *Seamless class booking*
- *Crowd meter (occupancy)*
- *Personalized workout tracking*
- *Locker and membership management*

- *Modal-based chatbot and onboarding flows*

2) Usability Principles Applied:

- **Progressive Disclosure:** Displaying essential information upfront while allowing users to explore advanced features as needed.
- **Feedback & Visibility:** Providing real-time notifications, confirmations, and visual cues for user actions (e.g., booking confirmations, payment processing).
- **Mobile-First Design:** Prioritizing a responsive, touch-friendly UI for an optimized mobile experience across iOS and Android devices.

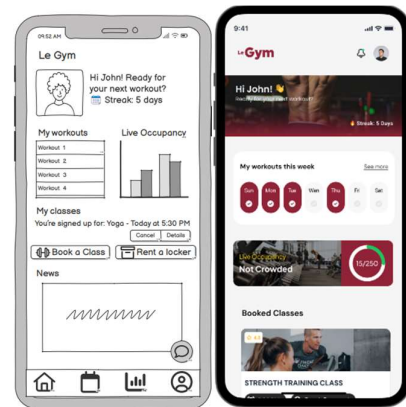


Figure 13: Prototype evolution

I. Deliver Phase

Once usability testing is complete, the next step is to deploy the final version of the application, ensuring high performance, reliability, and scalability. However, the launch of the app is not the end of development; usability refinement and feature improvements will be ongoing. By monitoring user interactions and gathering continuous feedback, we can identify and address pain points, optimize performance, and introduce new enhancements based on real-world usage.

1) Smooth Deployment

Before making the application available to users, we will conduct thorough quality assurance testing to ensure stability, security, and responsiveness. This includes:

- Finalizing all core functionalities (e.g., class booking, payment processing, workout tracking) to ensure smooth operation.

- Load testing the system to handle peak usage times without performance degradation.
- Ensuring full compatibility across mobile platforms (iOS and Android) and different device screen sizes.
- Deploying on cloud infrastructure for scalability and reliability, allowing the system to grow with user demand.

A well-planned deployment will minimize downtime, prevent unexpected errors, and provide a seamless onboarding experience for students using the app for the first time.

2) *Real-World Testing & Feedback Collection*

After deployment, monitoring user behavior and gathering feedback will be critical to identifying potential usability issues and areas for improvement. We will implement:

- In-app analytics to track how users interact with key features and identify potential friction points.
- Feedback forms and surveys within the app to collect direct user input on usability, design, and functionality.
- Bug reporting and support channels to allow users to report issues easily and receive assistance.
- Engagement tracking to determine which features drive the most value for users and which may need refinement.

By combining quantitative data (analytics) and qualitative insights (user feedback), we can make informed decisions about what needs improvement and what additional features may be valuable.

3) *Iterative Updates & Continuous Improvement*

The application will follow a continuous improvement model, with regular updates based on real-world usage and user needs. The development team will:

- Release periodic updates to fix bugs, enhance performance, and introduce new features.
- Optimize the UI and workflows based on user feedback and interaction trends.
- Expand personalization features, such as workout recommendations, progress tracking, and adaptive notifications, to enhance engagement.
- Improve accessibility and inclusivity, ensuring that the app remains easy to use for students of all backgrounds and fitness levels.

4) *Usability Principles Applied*

To ensure long-term success, the following usability principles will guide the post-launch updates and refinements:

- Continuous Improvement: Regular updates will ensure the app remains relevant, functional, and optimized based on user needs.
- User Empowerment: Personalization features, such as adaptive workout plans, goal tracking, and custom reminders, will help users stay engaged and motivated.

By following a data-driven, user-centered approach, the app will continue evolving to provide the best possible gym management experience for Concordia students. The commitment to ongoing refinement and enhancement ensures that *Le Gym* remains an efficient, accessible, and engaging fitness platform in the long run.

V. REQUIRED TECHNOLOGIES, TOOLS AND TEAM EXPERTISE

To successfully develop, test, and deploy the application, we utilized a combination of software frameworks, development tools, and testing methodologies to ensure a functional, efficient, and user-friendly gym management system. Each component plays a critical role in optimizing performance, security, and usability.

J. *Frontend Development*

The frontend of the application will be built using React, with React Native enabling cross-platform compatibility for both iOS and Android devices. By leveraging React's component-based architecture, we can ensure scalability, code reusability, and a consistent user experience across different platforms. The mobile-first approach will prioritize smooth navigation, intuitive interactions, and responsiveness, ensuring users can seamlessly book classes, track workouts, and access gym services on the go.

K. *Backend Infrastructure*

The backend will be powered by Spring Boot and MySQL, providing a secure, scalable, and high-performance environment for handling user data and processing transactions. Spring Boot is well-suited for managing key features such as authentication, class scheduling, and payment processing, ensuring a fast and reliable experience for students. MySQL will store and manage critical user data, including workout history, class bookings, and locker reservations, enabling real-time data retrieval while maintaining security and efficiency. This backend structure ensures stability and long-term maintainability, allowing the system to handle growing numbers of users without performance degradation.

L. *UI/UX Design*

Creating an intuitive and visually appealing user experience is crucial for the success of the application. Balsamiq and Figma were used to design wireframes, interactive prototypes, and polished UI layouts, ensuring that the application is aesthetically pleasing and easy to navigate. The UI will follow modern design principles, incorporating clear visual hierarchy, minimal cognitive load, and accessibility considerations to accommodate users with different fitness levels and technical proficiency. By continuously refining the interface based on user feedback, we will ensure a seamless experience that enhances usability and engagement.

M. *Testing & Debugging*

To maintain reliability, security, and performance, the application will undergo comprehensive testing using Jest. Jest will be used for unit testing, allowing us to validate core functionalities, such as user authentication, booking management, and payment processing. These testing

methodologies will help us identify and resolve bugs, usability issues, and performance bottlenecks before deployment, ensuring a smooth and error-free user experience.

By integrating these technologies and tools, the development team will work collaboratively to create a high-performing, user-friendly, and scalable gym management application. This project is not only aimed at enhancing the gym experience for Concordia students but also serves as an opportunity for hands-on learning in software development, UI/UX design, and usability testing. Our focus on efficiency, accessibility, and user-centered design will ensure the application meets students' fitness needs while providing a practical and valuable learning experience for the development team.

VI. BIGGEST RISKS & CHALLENGES

N. Data Accuracy Issues

Live occupancy tracking must provide real-time, reliable data to ensure students can make informed decisions about when to visit Le Gym. However, several challenges can impact data accuracy:

- If the system relies on manual check-ins or approximations, it may not correctly reflect the actual number of people inside the gym.
- Inaccurate occupancy data could result in unexpected overcrowding, frustration among users, and even safety concerns related to capacity limits.
- Ensuring accurate tracking requires integration with hardware-based solutions (e.g., RFID card scanners, Bluetooth beacons, or computer vision systems), which may introduce additional technical and financial constraints.

O. Technical Complexity

Developing and deploying a real-time occupancy tracking system is technically demanding:

- The system needs low-latency data updates to reflect real-time conditions without noticeable lag.
- It must seamlessly integrate with Le Gym's existing infrastructure, such as student ID scanners, access control systems, or sensor-based tracking.
- Developing an algorithm that accurately processes live occupancy changes while preventing data inconsistencies requires a significant development effort and specialized expertise.

P. Scalability and Performance Risks

The real-time nature of occupancy tracking means the system must handle multiple user queries simultaneously without slowing down the app.

- During peak hours, the database will be processing continuous read-and-write

operations, increasing the risk of system overload.

- The backend must be designed to scale efficiently to accommodate growing user demand, ensuring that thousands of users can access live occupancy data without performance degradation.

Q. Mitigation Strategy

To address these challenges and ensure the reliable and efficient tracking of live occupancy data, we plan to implement the following risk-mitigation strategies:

1) Leverage Existing Infrastructure

Instead of building a completely new tracking system from scratch, we will explore ways to integrate with Le Gym's existing check-in infrastructure (e.g., student ID scanners, access control systems).

- If students already scan their ID cards upon entry, we can use this data to estimate current gym occupancy levels without needing additional hardware.
- This approach reduces development complexity, ensures data consistency, and minimizes implementation costs.

2) Implement a Queued Update System

Rather than instantaneous data refreshes, we will use a queued update system that balances performance and accuracy.

- The system will process occupancy data updates in short intervals (e.g., every 30–60 seconds) instead of trying to reflect every single change in real time.
- This will help reduce the load on the backend, ensuring that the system can handle high user traffic without performance degradation.
- Users will still receive near-real-time information while avoiding potential data inconsistencies caused by multiple simultaneous updates.

3) Fallback Option: Manual Check-In System

If real-time tracking proves too complex or unreliable within the given constraints, we will fallback to a manual check-in and check-out system.

- Users will be required to scan their student ID at entry and exit, providing a basic but functional way to estimate the number of people in the gym.
- While this approach is less sophisticated, it still allows users to get a general idea of gym occupancy levels, helping them plan their workouts accordingly.
- Over time, this system can be enhanced with automation and real-time integration as more resources become available.

VII. RESULTS

R. Final Prototype and Implementation

The final prototype of the Le Gym application closely mirrors the high-fidelity Figma designs and integrates all core features defined during the earlier stages. Developed using React Native, the application provides a seamless and responsive experience on both iOS and Android devices. The app structure was progressively refined. Each prototype iteration allowed the team to validate user flows, identify potential friction points, and improve functionality before implementation.

S. How the Prototype Meets User Needs

The application successfully addresses the needs identified during the Discover and Define phases:

- *Beginner users* benefit from simplified navigation, class suggestions, and visual progress tracking.
- *Regular users* can manage their schedules and goals more efficiently.
- *Advanced users* gain access to real-time occupancy, advanced tracking, and flexible training tools.

The app also delivers on the six key user goals. These user-centric features were validated through usability principles such as visibility of system status, error prevention, and task efficiency.

T. Evaluation Insights and Improvement Opportunities

Although full-scale user testing is scheduled for future iterations, initial evaluations suggest:

- 1) *Strengths:*
 - Fast and intuitive booking workflows
 - Visual clarity and consistent UI design
 - Real-time feedback improves decision-making
- 2) *Areas for Improvement:*
 - Crowded screen layouts on small devices
 - Need for clearer indicators when multiple bookings conflict
 - Integration with third-party wearables remains pending due to API limitations

Planned improvements for post-deployment updates include enhancing adaptive notifications, introducing goal-tracking features, and refining accessibility options for users with disabilities.

VIII. CONCLUSION

This project set out to address widespread student frustrations with Le Gym's current fitness management systems at Concordia University. By applying the Double Diamond Design Model and User-Centered Design principles, we developed a fully functional mobile application that simplifies gym interactions, improves accessibility, and encourages student engagement in physical wellness. Throughout the project, we:

- Conducted user behavior research through reviews and system analysis
- Defined actionable goals and created user personas and journey maps
- Designed and iterated on wireframes and high-fidelity prototypes
- Successfully developed the application using modern frameworks
- Ensured that all interactions followed usability heuristics

The final product empowers students of all backgrounds and fitness levels to take control of their workout routines, save time, and engage more actively in fitness programs. It also serves as a valuable learning experience in applying human-computer interaction principles, prototyping tools, and mobile app development. Moving forward, the application is well-positioned for real-world deployment and future expansion.

IX. GROUP CONTRIBUTIONS

U. Andres Ardila – Design Process & Development

Contributed to the overall design strategy, ensuring usability heuristics and the Double Diamond Design Model were properly integrated.

V. Abdul Sameer Jani Syed – Introduction, User Tasks & Goals & Development

Worked on the introduction and user needs analysis, defining the problem statement, user goals, and key app functionalities.

W. Ting Lok Hui – Define Phase (User Experience & System Structure)& Development

Focused on the Define Phase, creating user personas, empathy maps, and user flows, while structuring core app features.

X. Oluwamosope Adeyemi – Risk Assessment, Prototyping & Development

Conducted risk analysis, identifying challenges with live occupancy tracking, proposing risk mitigation strategies, and ensuring technical feasibility.

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